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GAS INSULATED SWITCHGEAR HIGH-SPEED EARTHING SWITCH AND GAS INSULATED SWITCHGEAR HAVING SAME

Abstract

The present disclosure is directed to a high-speed earthing switch for a gas insulated switchgear comprising an enclosure, a cylinder member inside the enclosure and having ventilation holes formed in a lower portion and a cylinder hole formed in an end portion, and a piston member inserted into the cylinder member, having one end protruding outside through the cylinder hole, and moving forward and backward by receiving force from a drive unit to spray air inside the cylinder member toward a contact portion when opened, wherein a fixed plate having a first surface is disposed on a piston member and a movable plate is disposed on a second surface of the fixed plate facing an opposite direction to the cylinder hole, the earthing switch comprising an elastic support unit enabling the movable plate to be elastically pressed to be in close contact with the second surface of the fixed plate.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a high-speed earthing switch for a gas insulated switchgear and a gas insulated switchgear having the same, and more particularly, a high-speed earthing switch for a gas insulated switchgear that is capable of achieving a state where a movable plate is always in close contact with a fixed plate by using an elastic unit, so that the movable plate pushes a spring upward to exhaust internal pressure of a cylinder when the internal pressure of the cylinder becomes predetermined pressure or higher during a closing operation, so as to decrease speed reduction during the closing operation, while gas inside the cylinder is compressed during an opening operation at a side, which is opposite to a side thereof when closed, in the close contact state of the movable plate, and the compressed gas is ejected along a flow path of a mover, so as to quickly extinguish arc, and a gas insulated switchgear having the same.

BACKGROUND ART

[0002] Typically, a gas insulated switchgear (GIS) is an electrical device that is installed in a circuit between a power side and a load side of an electrical power system, to protect the electrical power system and a load device by blocking current safely when the circuit is artificially open or closed under normal current conditions or when a fault current such as a ground fault or short circuit occurs on the circuit.

[0003] The gas insulated switchgear (GIS) generally includes a bushing unit that receives power from a high-voltage power source, a gas circuit breaker (CB), a disconnect switch, an earthing switch, a moving part, a control unit, etc. Here, the earthing switch is installed at a portion of a circuit and serves to manually ground (earth) a main circuit during device maintenance and inspection and to remove current remaining in a conductor during the device maintenance or inspection.

[0004] FIG. 1 illustrates an internal structure of a gas insulated switchgear according to the related art.

[0005] It is shown that a disconnect switch/earthing switch (DS/ES) 2 and high-speed earthing switches (HSES) 5, 9, and 10 are installed inside an enclosure 1. A drive unit or operating unit 3 is disposed outside the enclosure 1.

[0006] The high-speed earthing switch is largely divided into a fixed part and a moving part.

[0007] A fixed part 5 is disposed on a conductor 4 located inside the enclosure 1. The fixed part 5 includes a fixed contact 5a and a fixed contact holder 5b (see FIG. 2).

[0008] The moving part includes a movable shaft 6 that is connected to the drive unit 3 and rotates, a connector link 7 that is crank-connected to the movable shaft 6, a movable contact holder 8a and a movable contact 8b that are fixedly installed inside the enclosure 1, a mover 8c that is connected to the connector link 7 and moves forward and backward to connect or separate the movable contact 8b and the fixed contact 5a, and the like. In addition, a bus bar 9 is disposed to connect the moving part and an earth terminal 10. The bus bar 9 may connect the movable contact holder 8a and the earth terminal 10.

[0009] In this way, the high-speed earthing switch is configured by including the fixed part and the moving part that is in contact with or separated from the fixed part. The high-speed earthing switch is generally installed at a power inlet and used for discharging current on a line. The high-speed earthing switch must have electrostatic induction current switching performance, electromagnetic

induction switching performance, closing capacity (E1 class twice), short-time electrical connection performance, and the like.

[0010] The performance of high-speed earthing switch according to the related art depends on opening and closing speed, and a blocking failure occurs even at high speed of the high-speed earthing switch depending on a type of insulating gas. For example, blocking in air is more likely to fail than blocking in SF₆ gas.

[0011] FIGS. 2 and 3 illustrate operations of a high-speed earthing switch of a gas insulated switchgear according the related art. FIG. 2 shows an open state and FIG. 3 shows an earthed state.

[0012] In the high-speed earthing switch of the gas-insulated switchgear according to the related art, the movable shaft **6** is rotated by power transmitted from the drive unit **3**, and the mover **8c** connected to the movable shaft **6** moves forward and backward to connect (close) an earth circuit (bring the movable contact and the fixed contact into contact with each other) or disconnect (open) the earth circuit (separate the movable contact and the fixed contact from each other).

[0013] In the closed state, current flows to external ground sequentially via the fixed contact holder **5b**, the fixed contact **5a**, the mover **8c**, the movable contact **8b**, the movable contact holder **8a**, the bus bar **9**, and the earth terminal **10**.

[0014] However, in the high-speed earthing switch of the gas-insulated switchgear according to the related art, an arcing time for which arc lasts increases depending on a type of insulating gas, resulting in a blocking failure.

[0015] Korean Patent Registration No. 10-2135381 discloses a blowing mechanism to quickly extinguish arc.

[0016] The Korean Patent Registration No. 10-2135381 discloses a configuration that includes an enclosure, a cylinder member that is installed inside the enclosure and has a ventilation hole formed in a lower portion along a circumferential surface, and a piston member that is inserted into the cylinder member and moves forward and backward by receiving force from a drive unit to spray air inside the cylinder member toward a contact part upon when opened.

[0017] Additionally, a fixed plate is installed in the middle of the piston member, and a moving plate is disposed on top of the fixed plate to be slidable along the piston member.

[0018] Referring to drawings of Korean Patent Registration No. 10-2135381, when a piston member **20** moves downward and is brought into contact with a fixed part **40**, current remaining in a first conductor **14** flows to outside sequentially via the fixed part **40**, the piston member **20**, a movable contact **52**, a moving part holder **51**, a bus bar **46**, and an earth terminal **45**.

[0019] Here, the blowing mechanism includes a movable shaft **17** (in addition, a mechanism for transmitting force from the drive unit to the piston member may be applied), a piston member **20**, a cylinder member **30**, a fixed plate **60**, and a movable plate **70**.

[0020] FIG. 4 is a diagram showing an example in which the related art high-speed earthing switch is installed in a vertical direction, and FIG. 5 is a diagram showing an example in which the related art high-speed earthing switch is installed in a horizontal direction.

[0021] Referring to FIGS. 4 and 5, when the high-speed earthing switch is installed vertically or horizontally depending on a GIS layout of a substation, the movable plate **70** of the blowing mechanism included inside the switch generates a gap without being in close contact with the fixed plate **60** upon an opening operation, thereby failing to achieve performance required for arc blocking.

[0022] In other words, the movable plate **70** is not fixed and is located downward due to its own weight. Thus, there may be no performance problem in a vertical installation structure. However, when the movable plate **70** is installed horizontally or on a lower portion upwardly, an accurate operation is not performed depending on a state in which the movable plate is located.

DISCLOSURE OF INVENTION

Technical Problem

[0023] In order to solve the above problems, one aspect of the present disclosure is to provide a

high-speed earthing switch for a gas insulated switchgear that is capable of achieving a state where a movable plate of a blowing mechanism is always in close contact with a fixed plate by using an elastic support unit, so that the movable plate pushes a spring upward to exhaust internal pressure of a cylinder when the internal pressure of the cylinder becomes predetermined pressure or higher during a closing operation, so as to decrease speed reduction during the closing operation, while gas inside the cylinder is compressed during an opening operation at a side, which is opposite to a side thereof when closed, in a close contact state of the movable plate, and the compressed gas is ejected along a flow path of a mover so as to quickly extinguish arc.

[0024] Another aspect of the present disclosure is to provide a high-speed earthing switch for a gas-insulated switchgear that is capable of providing freedom of installation without being restricted by an installation direction of the high-speed earthing switch for the gas-insulated switchgear.

[0025] The present disclosure is not limited to those aspects mentioned above and other aspects and advantages of the present disclosure that are not mentioned will be understood by the following description and more clearly understood by embodiments of the present disclosure. Additionally, it will be readily apparent that the aspects and advantages of the present disclosure can be realized by mechanisms indicated in the claims appended below and combinations thereof.

Solution to Problem

[0026] In order to achieve the above aspects, the present disclosure provides a high-speed earthing switch for a gas-insulated switchgear.

[0027] The high-speed earthing switch for the gas insulated switchgear, which includes an enclosure, a cylinder member installed inside the enclosure and having ventilation holes formed in a lower portion thereof along a circumferential surface and a cylinder hole formed in an end portion thereof, and a piston member inserted into the cylinder member, having one end protruding outside through the cylinder hole, and moving forward and backward by receiving force from a drive unit to spray air inside the cylinder member toward a contact portion when opened, wherein a fixed plate having a first surface facing the cylinder hole is disposed on a piston member and a movable plate is disposed on a second surface of the fixed plate facing an opposite direction to the cylinder hole to be slidable along the piston member inside the cylinder member, [0028] includes an elastic support unit disposed inside the cylinder member to enable the movable plate to be elastically pressed to be in close contact with the second surface of the fixed plate.

[0029] Here, the elastic support unit includes [0030] a support member and an elastic spring, [0031] the support member is formed in a ring shape and is fixed at a predetermined position on an outer circumference of the piston member spaced apart from the movable plate, and [0032] the elastic spring is preferably disposed between the support member and the movable plate to elastically support the support member and the movable plate.

[0033] The elastic spring is [0034] disposed to be wound around an outer circumference of the piston member.

[0035] The elastic spring is provided in plurality and disposed to surround the outer circumference of the piston member to elastically support the support member and the movable plate multiply at different positions, and [0036] the plurality of elastic springs have different outer diameters from one another.

[0037] In addition, the elastic support part [0038] includes fixed members and elastic springs, [0039] each of the fixed members has one end extending from the second surface of the fixed plate by a predetermined length, [0040] a stopping end is formed on another end of each of the fixed members, and [0041] the elastic springs are disposed on the fixed members, respectively, to elastically support the movable plate and the stopping ends between the movable plate and the stopping ends.

[0042] The elastic support unit includes a first elastic support unit and a second elastic support unit, [0043] the first elastic support unit [0044] includes a support member and a first elastic spring, [0045] the support member is formed in a ring shape and is fixed at a predetermined position on an

outer circumference of the piston member spaced apart from the movable plate, and [0046] the first elastic spring elastically supports the support member and the movable plate between the support member and the movable plate, [0047] the second elastic support unit [0048] includes fixed members and second elastic springs, [0049] each of the fixed members has one end extending from the second surface of the fixed plate by a predetermined length, [0050] a stopping end is formed on another end of each of the fixed members, and [0051] the second elastic springs are disposed on the fixed members, respectively, to elastically support the movable plate and the stopping ends between the movable plate and the stopping ends.

[0052] According to another embodiment, the present disclosure provides a gas insulated switchgear including the aforementioned high-speed earthing switch.

Advantageous Effects of Invention

[0053] According to the aspects, the present disclosure can achieve a state where a movable plate is always in close contact with a fixed plate by using an elastic unit. Accordingly, when internal pressure of a cylinder becomes predetermined pressure or higher during a closing operation, the movable plate pushes a spring upward to exhaust the internal pressure, so as to decrease speed reduction during the closing operation. On the other hand, gas inside the cylinder is compressed during an opening operation at a side of the cylinder, which is opposite to a side thereof when closed, in the close contact state of the movable plate, and the compressed gas is ejected along a flow path of a mover, so as to quickly extinguish arc.

[0054] In addition, the present disclosure can provide freedom of installation so as not to be restricted in an installation direction of a high-speed earthing switch of a gas insulated switchgear.

[0055] In addition to the above-described effects, specific effects of the present disclosure are described below while explaining specific details for carrying out the invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0056] FIG. 1 is a view illustrating an internal structure of a gas insulated switchgear according to the related art.

[0057] FIGS. 2 and 3 are views illustrating operations of a high-speed earthing switch of a gas insulated switchgear according the related art.

[0058] FIG. 4 is a diagram illustrating an example in which the related art high-speed earthing switch is installed vertically.

[0059] FIG. 5 is a diagram illustrating an example in which the related art high-speed earthing switch is installed horizontally.

[0060] FIG. 6 is a diagram illustrating a configuration of a high-speed earthing switch in an upward installation structure according to the present disclosure.

[0061] FIG. 7 is a diagram illustrating a first example of an elastic support unit according to the present disclosure.

[0062] FIG. 8 is a diagram illustrating a configuration of a high-speed earthing switch in a horizontal installation structure according to the present disclosure.

[0063] FIG. 9 is a diagram illustrating an example in which an elastic spring is provided in plurality according to the present disclosure.

[0064] FIG. 10 is a diagram illustrating a configuration of a high-speed earthing switch, to which a second example of an elastic support unit is applied, according to the present disclosure.

[0065] FIG. 11 is a view illustrating the elastic support unit of FIG. 10.

[0066] FIG. 12 is a diagram illustrating a configuration of a high-speed earthing switch in a horizontal installation structure according to the present disclosure.

[0067] FIG. 13 is a diagram illustrating a third example of an elastic support unit according to the

present disclosure.

MODE FOR THE INVENTION

[0068] The above-mentioned objects, features, and advantages will be described in detail later with reference to the attached drawings, so that those skilled in the art will be able to easily implement the technical idea of the present disclosure. In describing the present disclosure, if a detailed explanation for a related known function or construction is considered to unnecessarily divert the gist of the present disclosure, such explanation is omitted. Hereinafter, a preferred embodiment according to the present disclosure will be described with reference to the accompanying drawings. In the drawings, the same or equivalent components are provided with the same or equivalent reference numbers.

[0069] Hereinafter, it will be understood that when an element is referred to as being disposed “on an upper portion or top (or lower portion or bottom)” of another element or “on (or below)” the another element, the element can be disposed in contact with an upper surface (or lower surface) of the another element or intervening elements may also be present.

[0070] Additionally, it will be understood that when an element is described as being “connected,” “coupled,” or “accessed” to another element, the elements can be directly connected or accessed to each other, but the other element can be “interposed” between the elements or the elements can be “connected,” “coupled,” or “accessed” to each other through the other element.

[0071] Hereinafter, a high-speed earthing switch for a gas insulated switchgear and a gas insulated switchgear having the same according to the present disclosure will be described with reference to the accompanying drawings.

[0072] The gas insulated switchgear according to the present disclosure includes a high-speed earthing switch according to the present disclosure.

[0073] FIG. 6 is a diagram illustrating a configuration of a high-speed earthing switch in an upward installation structure according to the present disclosure.

[0074] Hereinafter, the configuration of the high-speed earthing switch according to the present disclosure will be described with reference to FIG. 6.

[0075] The high-speed earthing switch includes an enclosure (or outer chest) (not illustrated). A cylinder member **100** is disposed inside the enclosure. An internal space is defined in the cylinder member **100**.

[0076] A cylinder hole **110** is formed at the top of the cylinder member **100**. A curvature is formed on an inner circumference of an upper end of the cylinder hole **110**.

[0077] Ventilation holes **120** are formed at intervals along a circumferential surface of the cylinder member **100**. The ventilation holes **120** may be formed along the circumferential surface which is located below the cylinder hole **110**.

[0078] A piston member **200** having a predetermined length is disposed in the cylinder member **100**.

[0079] An upper end of the piston member **200** protrudes upward through the cylinder hole **110**.

[0080] The piston member **200** may move forward and backward by receiving force from a drive unit (not illustrated) to spray air inside the cylinder member **100** to a contact part when opened.

[0081] A step **210** is formed at a predetermined position on the piston member **200**.

[0082] A fixed plate **300** having a predetermined thickness is installed on the piston member **200**. A central hole **310** is formed in a center of the fixed plate **300**.

[0083] The central hole **310** of the fixed plate **300** is fitted to the piston member **200**. The fixed plate **300** is fixed by being caught on the step **210**.

[0084] The fixed plate **300** may alternatively be fixed to the piston member **200** in a welding manner. Also, a locking structure which includes a locking groove and a locking protrusion may be disposed between an outer circumference of the piston member **200** and an inner circumference of the central hole **310** of the fixed plate **300**.

[0085] In addition, a first surface of the fixed plate **300** is disposed on a side facing the cylinder

hole **110**, and a second surface of the fixed plate **300** is disposed on a side facing an opposite side to the cylinder hole **110**.

[0086] The outer circumference of the fixed plate **300** is disposed to slip on an inner circumference of the inner space of the cylinder member **100**. A groove which is spaced apart from the inner circumference of the cylinder member **100** may be formed in an outer circumferential surface of the fixed plate **300**.

[0087] A movable plate **400** having a predetermined outer diameter is disposed below the fixed plate **300**.

[0088] A through hole **410** is formed through a center of the movable plate **400**, and the piston member **200** is inserted through the through hole **410**. Accordingly, the movable plate **400** is disposed to be slidable along the piston member.

[0089] An upper surface of the movable plate **400** is disposed to be contactable with the second surface, which is the lower surface of the fixed plate **300**.

[0090] A thickness of the moving plate **400** may be thinner than a thickness of the fixed plate **300** by a predetermined value or less. Additionally, the movable plate **400** may have rigidity that is higher than that of the fixed plate **300** by a predetermined level or more.

[0091] The movable plate **400** according to the present disclosure may receive elastic force upward by an elastic support unit **500** and may be elastically pressed to be in contact with the second surface of the fixed plate **300**.

[0092] The elastic support unit **500** is disposed inside the cylinder member **100**.

[0093] FIG. **7** is a diagram illustrating a first example of an elastic support unit according to the present disclosure.

[0094] Referring to FIG. **7**, according to the first example, the elastic support unit **500** according to the present disclosure includes a support member **410** and an elastic spring **520**.

[0095] The support member **510** is formed in a ring shape and is fixed at a predetermined position on the outer circumference of the piston member **200**, which is spaced apart from the movable plate **400**.

[0096] The elastic spring **520** may elastically support the support member **510** and the movable plate **400** between them.

[0097] The elastic spring **520** is formed in a shape wound around the outer circumference of the piston member **200**.

[0098] Through this configuration, the movable plate **400** according to the present disclosure can be elastically pressed upward by elastic force of the elastic spring **520** supported on the support member **510**, which is installed on the outer circumference of the piston member **200**, so as to be always in close contact with the second surface of the fixed plate **300**.

[0099] In the present disclosure, as the state in which the movable plate **400** is always in close contact with the fixed plate **300** is realized by using the elastic support unit **500**, when internal pressure of the cylinder member **100** becomes predetermined pressure or higher during a closing operation, the movable plate **400** pushes the elastic spring **520** upward to exhaust the internal pressure, so as to decrease speed reduction during the closing operation. On the other hand, gas inside the cylinder member **100** is compressed during an opening operation at a side of the cylinder, which is opposite to a side of the cylinder when closed, and the compressed gas is ejected along a flow path of the mover, so as to quickly extinguish arc.

[0100] FIG. **8** is a diagram illustrating a configuration of a high-speed earthing switch in a horizontal installation structure according to the present disclosure.

[0101] In addition, as illustrated in FIG. **7**, even when the high-speed earthing switch according to the present disclosure is installed along the horizontal direction, the movable plate **400** is always forcibly in close contact with the second surface of the fixed plate **300**, so as to suppress a generation of a gap from the second surface of the fixed plate **300**, thereby effectively achieving performance required for arc blocking.

[0102] FIG. **9** is a diagram illustrating an example in which an elastic spring is provided in plurality according to the present disclosure.

[0103] Referring to FIG. **9**, the elastic spring **520** according to the present disclosure may be provided in plurality, which are disposed to surround the outer circumference of the piston member **200**.

[0104] The plurality of elastic springs **520** may have different outer diameters. Accordingly, the elastic springs **520** with the different outer diameters can elastically support the support member **510** and the movable plate **400** multiply at different positions.

[0105] In the movable plate **400** according to the present disclosure, by virtue of the elastic springs **520** with the different outer diameters installed at the multiple positions, even when one elastic spring decreases in elastic force, another elastic spring can apply elastic force to the movable plate **400**. This may result in more efficiently suppressing a generation of a gap between the movable plate **400** and the fixed plate **300**.

[0106] FIG. **10** is a diagram illustrating a configuration of a high-speed earthing switch, to which a second example of an elastic support unit is applied, according to the present disclosure. FIG. **11** is a view illustrating the elastic support unit of FIG. **10**.

[0107] Referring to FIGS. **10** and **11**, according to the second example, the elastic support unit **600** according to the present disclosure includes fixed members **610** and elastic springs **620**.

[0108] One end of each of the fixed members **610** extends from the second surface of the fixed plate **400** by a predetermined length.

[0109] A stopping end **611** is formed on another end of each of the fixed members **610**.

[0110] The elastic springs **620** may also be disposed on the fixed members **610**, respectively, to elastically support the movable plate **400** and the stopping ends **611** between them.

[0111] Through this configuration, the moving plate **400** according to the present disclosure can be elastically pressed upward by elastic forces of the plurality of elastic springs **620**, which are caught on the stopping ends **611** of the fixed members **610**, respectively, so as to be in a state where it is always in close contact with the second surface of the fixed plate.

[0112] FIG. **12** is a diagram illustrating a configuration of a high-speed earthing switch in a horizontal installation structure according to the present disclosure.

[0113] In addition, as illustrated in FIG. **12**, even when the high-speed earthing switch according to the present disclosure is installed along the horizontal direction, the movable plate **400** is always forcibly in close contact with the second surface of the fixed plate **300**, so as to suppress a generation of a gap from the second surface of the fixed plate **300**, thereby effectively achieving performance required for arc blocking.

[0114] FIG. **13** is a diagram illustrating a third example of an elastic support unit according to the present disclosure.

[0115] Referring to FIG. **13**, the elastic support unit according to the third example of the present disclosure includes a first elastic support unit **500** and a second elastic support unit **600**.

[0116] The first elastic support unit **500** includes a support member **510** and a first elastic spring **520**.

[0117] The support member **510** is formed in a ring shape and is fixed at a predetermined position on the outer circumference of the piston member **200**, which is spaced apart from the movable plate **400**.

[0118] The first elastic spring **520** elastically supports the support member **510** and the movable plate **400** between them.

[0119] The second elastic support unit **600** includes fixed members **610** and second elastic springs **620**.

[0120] One end of each of the fixed members **610** extends from the second surface of the fixed plate **300** by a predetermined length.

[0121] A stopping end **611** is formed on another end of each of the fixed members **610**.

[0122] The second elastic springs **620** may also be disposed on the fixed members **610**, respectively, to elastically support the movable plate **400** and the stopping ends **611** between them.

[0123] Through this, the movable plate **400** according to the present disclosure is elastically pressed in an upward direction by the first elastic support unit **500** through elasticity with the support member **510** installed at a predetermined position of the piston member **200**.

[0124] At the same time, the movable plate **400** is elastically pressed in a downward direction by the second elastic support unit **600** through elasticity with the fixed members **610** supported on the stopping ends **611** of the fixed members **610** extending from the fixed plate **300**.

[0125] Accordingly, since the movable plate **400** according to the present disclosure is elastically pressed at different positions and in different directions by the first and second elastic support units **500** and **600**, the movable plate **400** can be more efficiently in contact with the second surface of the fixed plate, thereby suppressing a generation of a gap (clearance).

[0126] Here, the positions of the first and second elastic support units **500** and **600** may be disposed contrarily to the above example, based on the central axis of the piston member **200**.

[0127] According to the above configuration and operation, the present disclosure can achieve a state where a movable plate is always in close contact with a fixed plate by using an elastic unit. Accordingly, when internal pressure of a cylinder becomes predetermined pressure or higher during a closing operation, the movable plate pushes a spring upward to exhaust the internal pressure, so as to decrease speed reduction during the closing operation. On the other hand, gas inside the cylinder is compressed during an opening operation at a side, which is opposite to a side thereof when closed, in the close contact state of the movable plate, and the compressed gas is ejected along a flow path of a mover, so as to quickly extinguish arc.

[0128] In addition, the present disclosure can provide freedom of installation so as not to be restricted in an installation direction of a high-speed earthing switch of a gas insulated switchgear.

[0129] The present disclosure may be subject to various substitutions, modifications, and changes by those skilled in the art, to which the present disclosure pertains, without departing from the technical idea of the present disclosure, and is not limited by the foregoing embodiment and accompanying drawings.

Claims

1. A high-speed earthing switch for a gas insulated switchgear, comprising an enclosure, a cylinder member installed inside the enclosure and having ventilation holes formed in a lower portion thereof along a circumferential surface and a cylinder hole formed in an end portion thereof, and a piston member inserted into the cylinder member, having one end protruding outside through the cylinder hole, and moving forward and backward by receiving force from a drive unit to spray air inside the cylinder member toward a contact portion when opened, wherein a fixed plate having a first surface facing the cylinder hole is disposed on a piston member and a movable plate is disposed on a second surface of the fixed plate facing an opposite direction to the cylinder hole to be slidable along the piston member inside the cylinder member, the high-speed earthing switch comprising: an elastic support unit disposed inside the cylinder member to enable the movable plate to be elastically pressed to be in close contact with the second surface of the fixed plate.
2. The high-speed earthing switch of claim 1, wherein the elastic support unit comprises a support member and an elastic spring, the support member is formed in a ring shape and is fixed at a predetermined position on an outer circumference of the piston member spaced apart from the movable plate, and the elastic spring is disposed between the support member and the movable plate to elastically support the support member and the movable plate.
3. The high-speed earthing switch of claim 2, wherein the elastic spring is disposed to be wound around an outer circumference of the piston member.
4. The high-speed earthing switch of claim 2, wherein the elastic spring is provided in plurality and

disposed to surround the outer circumference of the piston member, and the plurality of elastic springs have different outer diameters from one another.

5. The high-speed earthing switch of claim 1, wherein the elastic support unit comprises fixed members and elastic springs, each of the fixed members has one end extending from the second surface of the fixed plate by a predetermined length, a stopping end is formed on another end of each of the fixed members, and the elastic springs are disposed on the fixed members, respectively, to elastically support the movable plate and the stopping ends between the movable plate and the stopping ends.

6. The high-speed earthing switch of claim 1, wherein the elastic support unit includes a first elastic support unit and a second elastic support unit, the first elastic support unit comprises a support member and a first elastic spring, the support member is formed in a ring shape and is fixed at a predetermined position on an outer circumference of the piston member spaced apart from the movable plate, the first elastic spring elastically supports the support member and the movable plate between the support member and the movable plate, the second elastic support unit comprises fixed members and second elastic springs, each of the fixed members has one end extending from the second surface of the fixed plate by a predetermined length, a stopping end is formed on another end of each of the fixed members, and the second elastic springs are disposed on the fixed members, respectively, to elastically support the movable plate and the stopping ends between the movable plate and the stopping ends.

7. A gas insulated switchgear comprising the high-speed earthing switch of claim 1.
